

Median

Median is the value of the variable which divides the whole set of data into two equal parts. It is the value such that in a set of observations, 50% observations are above and 50% observations are below it. Hence the median is a positional average.

(a) Median for Ungrouped or Raw data:

In this case, the data is arranged in either ascending or descending order of magnitude.

(i) If the number of observations n is an odd number, then the median is represented by the numerical value of x , corresponds to the positioning point of $n+1 / 2$ in ordered observations. That is,

Median = value of $(n+1 / 2)^{\text{th}}$ observation in the data array

If the number of observations n is an even number, then the median is defined as the arithmetic mean of the middle values in the array That is,

$$\text{Median} = \frac{\text{value of } \left(\frac{n}{2}\right)^{\text{th}} \text{ observation} + \text{value of } \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2}$$

Example

The number of rooms in the seven five stars hotel in Chennai city is 71, 30, 61, 59, 31, 40 and 29. Find the median number of rooms

Solution:

Arrange the data in ascending order 29, 30, 31, 40, 59, 61, 71

$n = 7$ (odd)

Median = $7+1 / 2 = 4^{\text{th}}$ positional value

Median = 40 rooms

Example

The export of agricultural product in million dollars from a country during eight quarters in 1974 and 1975 was recorded as 29.7, 16.6, 2.3, 14.1, 36.6, 18.7, 3.5, 21.3

Find the median of the given set of values

Solution:

We arrange the data in descending order

36.6, 29.7, 21.3, 18.7, 16.6, 14.1, 3.5, 2.3

$$n = 8 \text{ (even)}$$

$$\begin{aligned}\text{Median} &= \frac{4^{\text{th}} \text{ item} + 5^{\text{th}} \text{ item}}{2} \\ &= \frac{18.7 + 16.6}{2} \\ &= 17.65 \text{ million dollars}\end{aligned}$$

Cumulative Frequency

In a grouped distribution, values are associated with frequencies. The cumulative frequencies are calculated to know the total number of items above or below a certain limit. This is obtained by adding the frequencies successively up to the required level. This cumulative frequencies are useful to calculate median, quartiles, deciles and percentiles.

(b) Median for Discrete grouped data

We can find median using following steps

- i. Calculate the cumulative frequencies
- ii. Find $(N+1)/2$, where $N = \Sigma f = \text{total frequencies}$
- iii. Identify the cumulative frequency just greater than $(N+1)/2$

iv. The value of x corresponding to that cumulative frequency is the $(N+1)/2$ median.

Example

The following data are the weights of students in a class. Find the median weights of the students

Weight(kg)	10	20	30	40	50	60	70
Number of Students	4	7	12	15	13	5	4

Solution:

Weight (kg) x	Frequency f	Cumulative Frequency $c.f$
10	4	4
20	7	11
30	12	23
40	15	38
50	13	51
60	5	56
70	4	60
Total	$N = 60$	

Here, $N = \sum f = 60$

$$\frac{N+1}{2} = 30.5$$

The cumulative frequency greater than 30.5 is 38. The value of x corresponding to 38 is 40. The median weight of the students is 40 kg.

(c) Median for Continuous grouped data

In this case, the data is given in the form of a frequency table with class-interval etc., The following formula is used to calculate the median.

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

Where

l = Lower limit of the median class

N = Total Numbers of frequencies

f = Frequency of the median class

m = Cumulative frequency of the class preceding the median class

c = the class interval of the median class.

From the formula, it is clear that one has to find the median class first. Median class is, that class which correspond to the cumulative frequency just greater than $N/2$.

Example

The following data attained from a garden records of certain period Calculate the median weight of the apple

Weight in grams	410 – 420	420 – 430	430 – 440	440 – 450	450 – 460	460 – 470	470 – 480
Number of apples	14	20	42	54	45	18	7

Solution:

Weight in grams	Number of apples	Cumulative Frequency
410 – 420	14	14
420 – 430	20	34
430 – 440	42	76
440 – 450	54	130
450 – 460	45	175
460 – 470	18	193
470 – 480	7	200
Total	N = 200	

$$\frac{N}{2} = \frac{200}{2} = 100.$$

Median class is 440 – 450

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

$$l = 440, \quad \frac{N}{2} = 100, \quad m = 76, \quad f = 54, \quad c = 10$$

$$\begin{aligned} \text{Median} &= 440 + \frac{100 - 76}{54} \times 10 \\ &= 440 + \frac{24}{54} \times 10 = 440 + 4.44 = 444.44 \end{aligned}$$

The median weight of the apple is 444.44 grams

Example

The following table shows age distribution of persons in a particular region:

Age (years)	No. of persons (in thousands)
Below 10	2
Below 20	5
Below 30	9
Below 40	12
Below 50	14
Below 60	15
Below 70	15.5
Below 80	15.6

Find the median age.

Solution:

We are given upper limit and less than cumulative frequencies. First find the class-intervals and the frequencies. Since the values are increasing by 10, hence the width of the class interval is equal to 10.

Age groups	No. of persons (in thousands) f	cf
0 – 10	2	2
10 – 20	3	5
20 – 30	4	9
30 – 40	3	12
40 – 50	2	14
50 – 60	1	15
60 – 70	0.5	15.5
70 – 80	0.1	15.6
Total	N = 15.6	

$$\left(\frac{N}{2}\right) = \frac{15.6}{2} = 7.8$$

Median lies in the 20 – 30 age group

$$\begin{aligned}\text{Median} &= l + \frac{\frac{N}{2} - m}{f} \times c \\ &= 20 + \frac{7.8 - 5}{4} \times 10\end{aligned}$$

$$\text{Median} = 27 \text{ years}$$

Example

The following is the marks obtained by 140 students in a college. Find the median marks

Marks	Number of students
10-19	7
20-29	15
30-39	18
40-49	25
50-59	30
60-69	20
70-79	16
80-89	7
90-99	2

Solution:

Class boundaries	f	Cf
9.5 -19.5	7	7
19.5-29.5	15	22
29.5- 39.5	18	40
39.5-49.5	25	65
49.5-59.5	30	95
59.5-69.5	20	115
69.5-79.5	16	131
79.5-89.5	7	138
89.5-99.5	2	140
Total	N =140	

$$\text{Median} = l + \left(\frac{\frac{N}{2} - m}{f} \right) \times c$$

$$\frac{N}{2} = \frac{140}{2} = 70$$

Here $l = 49.5$, $f = 30$, $m = 65$, $c = 10$

$$\begin{aligned} \text{Median} &= 49.5 + \left(\frac{70 - 65}{30} \right) \times 10 \\ &= 49.5 + 1.67 \\ &= 51.17 \end{aligned}$$

Graphical method for Location of median

Median can be located with the help of the cumulative frequency curve or 'ogive'.

The procedure for locating median in a grouped data is as follows:

Step 1 : The class intervals, are represented on the horizontal axis (x-axis)

Step 2 : The cumulative frequency corresponding to different classes is calculated. These cumulative frequencies are plotted on the vertical axis (y-axis) against the upper limit of the respective class interval

Step 3 : The curve obtained by joining the points by means of freehand is called the '*less than ogive*'.

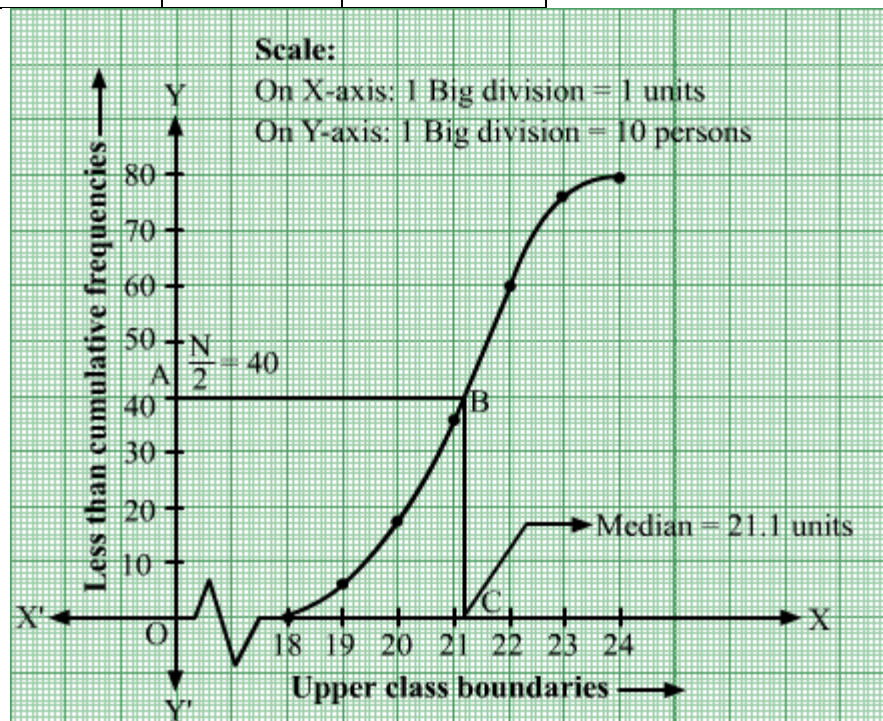
Step 4 : A horizontal straight line is drawn from the value $N/2$ and $N+1 / 2$ on the y-axis parallel to x- axis to meet the ogive. (depending on N is odd or even)

Step 5 : From the point of intersection, draw a line, perpendicular to the horizontal axis which meet the x axis at C say.

Step 6 : The value B joint the x axis which point C, gives the value of the median.

Example : Find median graphically.

Class interval	frequency	Cumulative frequency
17-18	0	0
18-19	5	5
19-20	14	19
20-21	20	39
21-22	21	60
22-23	15	75
23-24	5	80
Total	N=80	



$$N/2 = 80/2 = 40$$

$$\text{Medan} = 21 + \left(\frac{40-39}{21}\right) \times 1 = 21 + .047 = 21.047$$

Merits

- It is easy to compute. It can be calculated by mere inspection and by the graphical method
- It is not affected by extreme values.
- It can be easily located even if the class intervals in the series are unequal

Limitations

- It is not amenable to further algebraic treatment
- It is a positional average and is based on the middle item
- It does not take into account the actual values of the items in the serie

Partition Measures

Quartiles

A measure which divides an array into four equal parts is known as quartiles.

Each portion contains equal number of items. The first, second and third points are termed as first quartile (Q_1), second quartile (Q_2) (better named as median) and the third quartile (Q_3). The first quartile (Q_1) or lower quartile, has 25% of the items of the distribution below it and 75% of the items are greater than it. Q_2 (median) the second quartile or median has 50 % of the observations above it and 50% of the observations below it. The upper quartile or third quartile(Q_3) has 75% of the items of the distribution below it and 25% of the items are above it. Similarly the other two positional measures can be defined.

There are three quartiles denoted by Q_1 , Q_2 and Q_3 divides the frequency distribution in to four equal parts

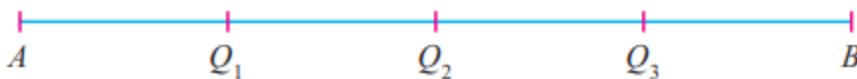


Fig.

That is 25 percent of data will lie below Q_1 , 50 percent of data below Q_2 and 75 percent below Q_3 . Here Q_2 is called the Median. Quartiles are obtained in almost the same way as median

Quartiles for Raw or Ungrouped data:

If the data set consist of n items and arranged in ascending order then

$$Q_1 = \left(\frac{n+1}{4} \right)^{th} \text{ item,}$$

$$Q_2 = \left(\frac{n+1}{2} \right)^{th} \text{ item and}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{th} \text{ item}$$

Example

Compute Q_1 and Q_3 for the data relating to the marks of 8 students in an examination given below 25, 48, 32, 52, 21, 64, 29, 57

Solution:

$$n = 8$$

Arrange the values in ascending order

21, 25, 29, 32, 48, 52, 57, 64 we have

$$\begin{aligned}Q_1 &= \left(\frac{n+1}{4}\right)^{th} \text{ item} \\&= \left(\frac{8+1}{4}\right)^{th} \text{ item} \\&= 2.25^{th} \text{ item} \\&= 2^{nd} \text{ item} + \left(\frac{1}{4}\right) (3^{rd} \text{ item} - 2^{nd} \text{ item}) \\&= 25 + 0.25 (29 - 25) \\&= 25 + 1.0\end{aligned}$$

$$Q_1 = 26$$

$$\begin{aligned}Q_3 &= 3\left(\frac{n+1}{4}\right)^{th} \text{ item} \\&= 3 \times (2.25)^{th} \text{ item} \\&= 6.75^{th} \text{ item} \\&= 6^{th} \text{ item} + 0.75 (7^{th} \text{ item} - 6^{th} \text{ item}) \\&= 52 + (0.75) (57 - 52) \\&= 52 + 3.75\end{aligned}$$

$$=52 + (57-52)=52+5=57$$

Quartiles for Discrete Series (grouped data)

Step 1: Find cumulative frequencies

Step 2 : Find $((N+1)/ 4)$

Step 3 : See in the cumulative frequencies, the value just greater than $((N+1)/ 4)$ the corresponding value of x is Q_1

Step 4 : Find $3((N+1)/ 4)$

Step 5 : See in the cumulative frequencies, the value just greater than $3((N+1)/ 4)$ then the corresponding value of x is Q_3 .

Example

Compute Q_1 and Q_3 for the data relating to age in years of 543 members in a village

Age in years	20	30	40	50	60	70	80
No. of members	3	61	132	153	140	51	3

Solution:

x	f	cf
20	3	3
30	61	64
40	132	196
50	153	349
60	140	489
70	51	540
80	3	543

$$\begin{aligned}
 Q_1 &= \left(\frac{N+1}{4} \right)^{th} \text{ item} \\
 &= \left(\frac{543+1}{4} \right)^{th} \text{ item} \\
 &= 136^{th} \text{ item}
 \end{aligned}$$

$$Q_1 = 40 \text{ years}$$

$$\begin{aligned}
 Q_3 &= 3 \left(\frac{N+1}{4} \right)^{th} \text{ item} \\
 &= 3 \left(\frac{543+1}{4} \right)^{th} \text{ item} \\
 &= 3 \times 136^{th} \text{ item} \\
 &= 408^{th} \text{ item}
 \end{aligned}$$

$$Q_3 = 60 \text{ years}$$

Quartiles for Continuous series (grouped data)

Step 1: Find cumulative frequencies

Step 2 : Find $(N/4)$

Step 3 : Q_1 class is the class interval corresponding to the value of the cumulative frequency just greater than $(N/4)$

Step 4 : Q_3 class is the class interval corresponding to the value of the cumulative frequency just greater than $3(N/4)$

$$Q_1 = l_1 + \frac{\frac{N}{4} - m_1}{f_1} \times c_1 \quad \text{and} \quad Q_3 = l_3 + \frac{3\left(\frac{N}{4}\right) - m_3}{f_3} \times C_3$$

where $N = \sum f$ = total of all frequency values

l_1 = lower limit of the first quartile class

f_1 = frequency of the first quartile class

c_1 = width of the first quartile class

m_1 = c.f. preceding the first quartile class

l_3 = lower limit of the 3rd quartile class

f_3 = frequency of the 3rd quartile class

m_3 = c.f. preceding the 3rd quartile class

c_3 = width of the third quartile class

Example

Calculate the quartiles Q_1 and Q_3 for wages of the labours given below

Wages (Rs.)	30-32	32-34	34-36	36-38	38-40	40-42	42-44
Labourers	12	18	16	14	12	8	6

Solution:

x	f	cf
30 – 32	12	12
32 – 34	18	30
34 – 36	16	46
36 – 38	14	60
38 – 40	12	72
40 – 42	8	80
42 – 44	6	86
	86	

$$\frac{N}{4} = \frac{86}{4} = 21.5$$

lies in the group 32 – 34

$$\begin{aligned}
 Q_1 &= l_1 + \frac{\frac{N}{4} - m_1}{f_1} \times c_1 \\
 &= 32 + \frac{21.5 - 12}{18} \times 2 \\
 &= 32 + \frac{19}{18} = 32 + 1.06 \\
 &= \text{Rs. } 33.06
 \end{aligned}$$

$$\frac{3N}{4} = \frac{3 \times 86}{4} = 64.5$$

$\therefore Q_3$ lies in the group 38 – 40

$$\begin{aligned}
 Q_3 &= l_3 + \frac{\frac{3N}{4} - m_3}{f_3} \times c_3 \\
 &= 38 + \frac{64.5 - 60}{12} \times 2 \\
 &= 38 + \frac{4.5}{12} \times 2 \\
 &= 38 + 0.75 = \text{Rs. } 38.75
 \end{aligned}$$

Deciles

A measure which divides an array into ten equal parts is known as deciles.

That is Deciles are the values which divides the series into ten equal parts. We get nine dividing positions namely $D_1, D_2, \dots, D_9$ which are called as deciles. Therefore there are nine deciles. It is to be noted that D_5 is equal to median.

Deciles are similar to quartiles. Quartiles divides ungrouped data into four quarters and Deciles divide data into 10 equal parts .

Example

Find the D_6 for the following data

11, 25, 20, 15, 24, 28, 19, 21

Solution:

Arrange in an ascending order

11, 15, 19, 20, 21, 24, 25, 28

$$\begin{aligned}
 D_6 &= \left(\frac{6(n+1)}{10} \right)^{th} \text{ item} \\
 &= \left(\frac{6(8+1)}{10} \right)^{th} \text{ item} \\
 &= \left(\frac{6(9)}{10} \right)^{th} \text{ item} \\
 &= [5.4]^{th} \text{ item} \\
 &= 5^{th} \text{ item} + (0.4) (6^{th} \text{ item} - 5^{th} \text{ item})
 \end{aligned}$$

$$\begin{aligned}
 D_6 &= 21 + (0.4)(24 - 21) \\
 &= 21 + (0.4)(3) \\
 &= 21 + 1.2 \\
 &= 22.2
 \end{aligned}$$

Example 5.34

Calculate D_5 for the frequency distribution of monthly income of workers in a factory

Income (in thousands)	0 – 4	4 – 8	8 – 12	12 – 16	16 – 20	20 – 24	24 – 28	28 – 32
No of persons	10	12	8	7	5	8	4	6

Solution:

Class	f	cf
0-4	10	10
4-8	12	22
8-12	8	30
12-16	7	37
16-20	5	42
20-24	8	50
24-28	4	54
28-32	6	60
	N=60	

$$\begin{aligned}
 D_5 &= \left(\frac{5N}{10} \right)^{\text{th}} \text{ item} \\
 &= \left(\frac{5(60)}{10} \right)^{\text{th}} \text{ item} \\
 &= 30^{\text{th}} \text{ item}
 \end{aligned}$$

This item in the interval 8-12

$$l = 8, \quad m = 22, \quad f = 8, \quad c = 4, \quad N = 60$$

$$\begin{aligned}
 D_5 &= l + \left(\frac{\frac{5N}{10} - m}{f} \right) \times c \\
 &= 8 + \left(\frac{30 - 22}{8} \right) \times 4 \\
 &= 8 + \frac{8}{8} \times 4 \\
 D_5 &= 12
 \end{aligned}$$

Percentiles

A measure which divides an array into hundred equal parts is known as Percentiles.

That is percentiles are the values which divides the series into hundred equal parts. We get 99 dividing positions $P_1, P_2, \dots, P_{99}$ which are called as percentiles. Therefore there are 99 percentiles. It is to be noted that P_{50} is equal to median.

The percentile values divide the frequency distribution into 100 parts each containing 1 percent of the cases. It is clear from the definition of quartiles, deciles and percentiles

Relationship

$$P_{25} = Q_1$$

$$P_{50} = \text{Median} = Q_2$$

$$P_{75} = \text{3rd quartile} = Q_3$$

Example

The following is the monthly income (in 1000) of 8 persons working in a factory. Find P_{30} income value

10,14, 36, 25, 15, 21, 29, 17

Solutions:

Arrange the data in an ascending order.

$$n = 8$$

10,14, 15,17,21,25, 29,36

$$\begin{aligned}
 P_{30} &= \left(\frac{30(n+1)}{100} \right)^{\text{th}} \text{ item} \\
 &= \left(\frac{30 \times 9}{100} \right)^{\text{th}} \text{ item} \\
 &= 2.7^{\text{th}} \text{ item} \\
 &= 2^{\text{nd}} \text{ item} + 0.7 (3^{\text{rd}} \text{ items} - 2^{\text{nd}} \text{ items}) \\
 &= 14 + 0.7(15 - 14) \\
 &= 14 + 0.7 \\
 P_{30} &= 14.7
 \end{aligned}$$

Example

Calculate P_{61} for the following data relating to the height of the plants in a garden

Heights (in cm)	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
No of plants	18	20	36	40	26	16

Solution:

<i>Class</i>	<i>f</i>	<i>cf</i>
0 – 5	18	18
5 – 10	20	38
10 – 15	36	74
15 – 20	40	114
20 – 25	26	140
25 – 30	16	156
	N=156	

$$\frac{61N}{100} = \left(\frac{61 \times 156}{100} \right)$$

$$= 95.16 \text{ item}$$

This item is the interval 15-20. Thus

$$l = 15, m = 74, f = 40, c = 5$$

$$P_{61} = l + \left(\frac{\frac{61N}{100} - m}{f} \right) \times c$$

$$= 15 + \left(\frac{95.16 - 74}{40} \right) \times 5$$

$$= 15 + \frac{21.16}{40} \times 5$$

$$= 17.645$$

$$P_{61} = 17.645$$

Mode

According to Croxton and Cowden, 'The mode of a distribution is the value at the point around which the items tend to be most heavily concentrated.'

In a busy road, where we take a survey on the vehicle - traffic on the road at a place at a particular period of time, we observe the number of two wheelers is more than cars, buses and other vehicles. Because of the higher frequency, we say that the modal value of this survey is 'two wheelers'

Mode is defined as the value which occurs most frequently in a data set. The mode obtained may be two or more in frequency distribution.

Computation of mode:

(a) For Ungrouped or Raw Data:

The mode is defined as the value which occurs frequently in a data set

Example 1

The following are the marks scored by 20 students in the class. Find the mode 90, 70, 50, 30, 40, 86, 65, 73, 68, 90, 90, 10, 73, 25, 35, 88, 67, 80, 74, 46(unimodal)

Solution:

Since the marks 90 occurs the maximum number of times, three times compared with the other numbers, mode is 90.

Example 2: A doctor who checked 9 patients' sugar level is given below. Find the mode value of the sugar levels. 80, 112, 110, 115, 124, 130, 100, 90, 150, 180

Solution:

Since each values occurs only once, there is no mode.

Example 3 :

Compute mode value for the following observations.

7, 10, 12, 10, 19, 2, 11, 3, 12(bimodal)

Solution:

Here, the observations 10 and 12 occurs twice in the data set, the modes are 10 and 12.

For discrete frequency distribution, mode is the value of the variable corresponding to the maximum frequency.

7, 3, 7, 9, 7, 3, 4, 6, 5, 3, 4, 9, 4 mode 7, 4, 3(multimodal)

Example 4:

Calculate the mode from the following data

Days of Confinement	6	7	8	9	10
Number of patients	4	6	7	5	3

Solution:

Here, 7 is the maximum frequency, hence the value of x corresponding to 7 is 8.

Therefore 8 is the mode.

(b) Mode for Continuous data:

The mode or modal value of the distribution is that value of the variate for which the frequency is maximum. It is the value around which the items or observations tend to be most heavily concentrated. The mode is computed by the formula.

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times c$$

Modal class is the class which has maximum frequency.

f_1 = frequency of the modal class

f_0 = frequency of the pre the modal class

f_2 = frequency of the post the modal class

c = width of the class limits

or

$$Mode = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i$$

Where, L= Lower limit of the modal class

$\Delta_1 = f_1 - f_0$ The difference between frequency of the modal class and pre modal class

$\Delta_2 = f_1 - f_2$ The difference between frequency of the modal class and post modal class

i = Class interval of the modal class.

Remarks

- (i) If $(2f_1 - f_0 - f_2)$ comes out to be zero, then mode is obtained by the following formula taking absolute differences $M_0 = l + \left(\frac{(f_1 - f_0)}{|f_1 - f_0| + |f_1 - f_2|} \times C \right)$
- (ii) If mode lies in the first class interval, then f_0 is taken as zero.
- (iii) The computation of mode poses problem when the modal value lies in the open-ended class.

Example 5 : The following data relates to the daily income of families in an urban area. Find the modal income of the families.

Income (`)	0-100	100-200	200-300	300-400	400-500	500-600	600-700
No.of persons	5	7	12	18	16	10	5

Solution:

Income (`)	No.of persons (f)
0-100	5
100-200	7
200-300	12 f_0
300-400	18 f_1
400-500	16 f_2
500-600	10
600-700	5

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times C$$

The highest frequency is 18, the modal class is 300-400

Here, $l = 300$, $f_0 = 12$, $f_1 = 18$, $f_2 = 16$,

$$\begin{aligned}
 \text{Mode} &= 300 + \frac{18 - 12}{2 \times 18 - 12 - 16} \times 100 \\
 &= 300 + \frac{6}{36 - 28} \times 100 \\
 &= 300 + \frac{6}{8} \times 100 \\
 &= 300 + \frac{600}{8} = 300 + 75 = 375
 \end{aligned}$$

The modal income of the families is 375.

Example 6:

The following table gives the daily wages (in taka) of a certain tea garden:

Daily wages	30-32	32-34	34-36	36-38	38-40	40-42	42-44	44-46	46-48	48-50
No. of workers	2	9	25	30	49	62	39	20	11	3

Determine modal wages.

Solution:

Calculation of Mode	
Weekly wage	f
30 – 32	2
32 – 34	9
34 – 36	25
36 – 38	30
38 – 40	49
40 – 42	f ₀
42 – 44	62 f ₁
44 – 46	39 f ₂
46 – 48	20
48 – 50	11
	3
	N = 250

For highest frequency 62, mode lies in the class 40 – 42. $f_1 - f_0 = 62 - 49 = 13$, $f_1 - f_2 = 62 - 39 = 29$

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 40 + \frac{13}{13 + 23} \times 2 = 40.72 \quad [\text{Ans.}]$$

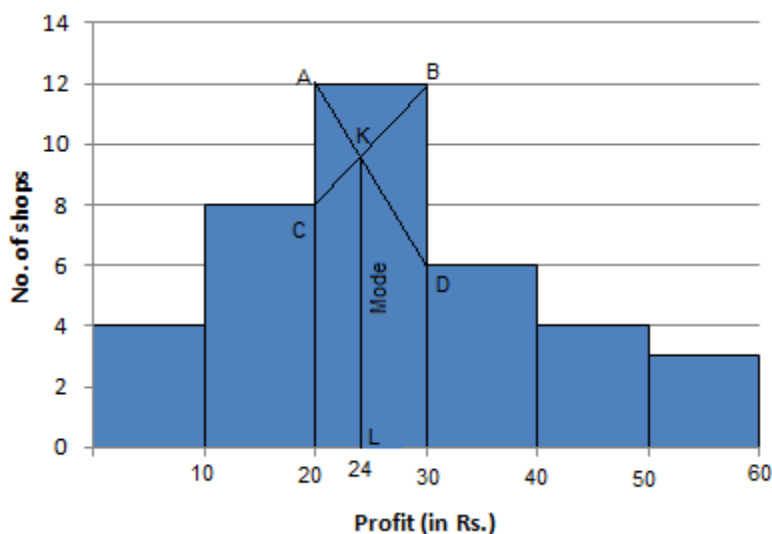
(c) Graphical Location of Mode

The following are the steps to locate mode by graph

- Draw a histogram of the given distribution.
- Join the rectangle corner of the highest rectangle (modal class rectangle) by a straight line to the top right corner (B) of the preceding rectangle(C). Similarly the top left corner (A) of the highest rectangle is joined to the top left corner (D) of the rectangle on the right.
- From the point of intersection of these two diagonal lines (k), draw a perpendicular line to the x –axis which meets at L.
- The value of x coordinate of L is the mode.

Example :Find mode graphically

Profits(in thous)	0-10	10-20	20-30	30-40	40-50	50-60
No.of shops	4	8 f ₀	12 f ₁	6 f ₂	4	3



For highest frequency 12, mode lies in the class 20-30

$$f_1 - f_0 = 12 - 8 = 4, f_1 - f_2 = 12 - 6 = 6$$

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 20 + \frac{4}{4 + 6} \times 10 = 20 + 4 = 24 \quad [\text{Ans.}]$$

Merits of Mode:

- ☐ It is comparatively easy to understand.
- ☐ It can be found graphically.
- ☐ It is easy to locate in some cases by inspection.
- ☐ It is not affected by extreme values.
- ☐ It is the simplest descriptive measure of average.

Demerits of Mode:

- ☐ It is not suitable for further mathematical treatment.
- ☐ It is an unstable measure as it is affected more by sampling fluctuations.
- ☐ Mode for the series with unequal class intervals cannot be calculated.
- ☐ In a bimodal distribution, there are two modal classes and it is difficult to determine the values of the mode.

Others related measurements

Central 50%:

Definition: Central 50% is the difference value between the first quartile value and the third quartile value of a data set.

Formula: Central 50% = $Q_3 - Q_1$

Five Number Summary :

Definition: The combination of smallest observation, first quartile, median, third quartile and largest observation is called five number summary.

Elements: Five number summary has five elements. i.e.,

- (1) The median of the data (Med.);(Q₂)
- (2) The lower quartile (Q₁);
- (3) The upper quartile (Q₃);
- (4) The smallest observation (X_{smallest});
- (5) The largest observation (X_{largest}).

Formula: Five Number Summary = X_{smallest} + Q₁ + Median + Q₃ + X_{largest}.

Example

The following data gives the Number of students studying in XI standard in 10 different schools 89,47,164 ,296,30,215,138,78,48,39 construct a five number summary (box plot) for the data.

Solution:

Step 1 : Arrange the data in order

30, 39, 47,48,78,89,138,164,215,296

Step 2 : Find the median

$$\text{Median} = \frac{78 + 89}{2} = 83.5$$

Step 3: Find Q₁

30,39,47,48,78

$$Q_1 = \frac{47 + 48}{2} = 47.5$$

Step 4: Find Q₃

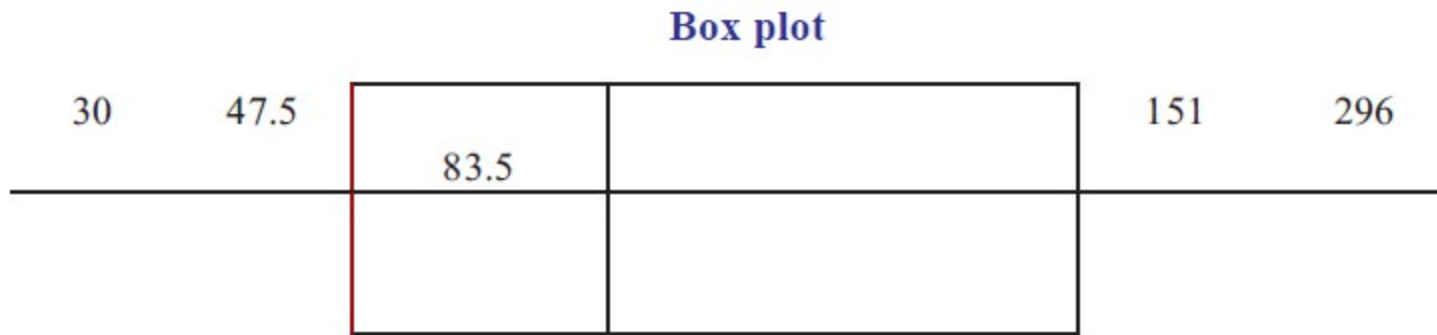
89,138,164,215,296

$$Q_3 = \frac{138 + 164}{2} = \frac{302}{2} = 151$$

Step 5 : Find the minimum(30) and maximum(296) values.

Step 6 : Locate the lowest value, Q₁, median, Q₃ and the highest value on the scale.

Step 7 : Draw a box through Q_1 and Q_3



Imperial Relationship among mean, median and mode:

A distribution in which the values of mean, median and mode coincide is known as symmetrical distribution.

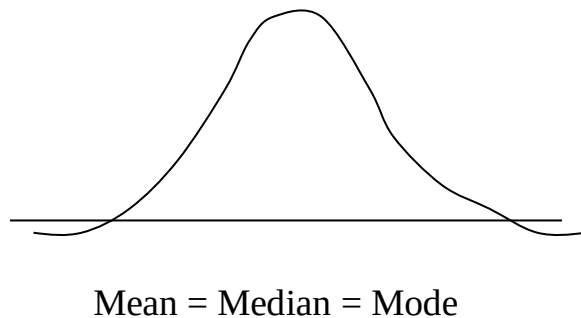


Figure : Relationship of arithmetic mean, median and mode.

Conversely stated, when the values of mean, median, and mode are not equal the distribution is known as symmetrical or skewed. In moderately (asymmetrical) skewed or symmetrical distributions a very important relationship exists among mean, median and mode. In such distribution the distance between the mean and median is approximately one-third of the distance between the mean and mode as will be clear from the following diagram

Karl Pearson has expressed this relationship as follows:

$$\text{Mean} - \text{Median} = \frac{1}{3} (\text{Mean} - \text{Mode})$$

$$\Rightarrow \text{Mode} = 3 \text{ Median} - 2 \text{ Means}$$

$$\Rightarrow \text{Median} = \text{Mode} + \frac{2}{3} (\text{Mean} - \text{Mode})$$

$$\Rightarrow \text{Mean} = \frac{2}{3} \text{Median} - \frac{1}{2} \text{Mode}$$

$$\Rightarrow \text{Mean} = \frac{3\text{Median} - \text{Mode}}{2}$$

If we know any of the two values out of the three here. We can compute the third from this relationship.

Example:

If the mean and median of a moderately asymmetrical series are 26.8 and 27.9 respectively, what would be its most probable mode?

Solution:

Using the empirical formula

$$\text{Mode} = 3 \text{ median} - 2 \text{ mean}$$

$$= 3 \times 27.9 - 2 \times 26.8$$

$$= 30.1$$

Problem 1

From the following frequency distribution, calculate median and mode. Of these two, which one do you think better? Why?

Frequency Distribution Table

Class	Frequency (f)
10 – 19.5	8
20 – 29.5	15
30 – 39.5	23
40 – 49.5	37
50 – 59.5	46
60 – 69.5	52
70 – 79.5	84

80 – 89.5	97
90 – 99.5	16
100 or over	5
	N = 383

Solution:

Calculation of Median and Mode

Class	f	c.f.
10 – 19.5	8	8
20 – 29.5	15	23
30 – 39.5	23	46
40 – 49.5	37	83
50 – 59.5	46	129
60 – 69.5	52	181
70 – 79.5	84	265
80 – 89.5	97	362
90 – 99.5	16	378
100 or over	5	383
	N = 383	

Median = Size of $\frac{N}{2}$ th item = $\frac{383}{2} = 191.5$ th item, which lies in the class 70 – 79.5. But the real class limit is 69.75 – 79.75

$$\text{Median} = L + \frac{\frac{N}{2} - \text{pcf}}{f} \times i = 69.75 + \frac{191.5 - 181}{84} \times 10 = 71 \quad [\text{Ans.}]$$

For highest frequency 97, mode lies in the class 80 – 89.5. But the real class limit is 79.75 – 89.75

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 79.75 + \frac{13}{13 + 81} \times 10 = 81.13 \quad [\text{Ans.}]$$

Comment: Of these two, we think median is better, because it is the more centered value than mode.

Problem 2

The following table gives the daily wages distribution of the employees of a commercial organization:

Daily wages (Tk) No. of employees

250 – 300	25
300 – 350	30
350 – 400	55
450 – 500	67
500 – 550	39
550 – 600	14

- (i) Compute arithmetic mean and median of the distribution.
- (ii) Find the wage range of the central 50% employees.
- (iii) It is decided to collect tax from those having income of Tk 530 and above. What percentage of the employees is taxable?

Solution:

Working note: Since the continuous frequency distribution has no class interval gap, so we should correct the frequency distribution table as follows:

Corrected Frequency Distribution Table

Class	Midpoint (x)	f	fx	cf
250 – 300	275	25	6875	25
300 – 350	325	30	9750	55
350 – 400	375	55	20625	110
400 – 450	425	0	0	110
450 – 500	475	67	31825	177
500 – 550	525	39	20475	216

550 – 600	575	14	8050	230
		N = 230	N = 97600	

(i) Arithmetic mean $(\bar{X}) = \frac{\Sigma fx}{N} = \frac{97600}{230} = 424.35$ [Ans.]

Median = $\frac{N}{2} = \frac{230}{2} = 115$ th item, which lies in the class 450-500.

$$\therefore \text{Median} = L + \frac{\frac{N}{2} - \text{p.c.f.}}{f} \times c = 450 + \frac{115 - 110}{67} \times 50 = 453.73 \text{ [Ans.]}$$

(ii) We know, Central 50% = Q_1 to Q_3

$Q_1 = \frac{N}{4} \times 1 = \frac{230}{4} \times 1 = 57.5$ th item, which lies in the class 350-400.

$$\therefore Q_1 = L + \frac{\frac{N}{4} \times 1 - \text{p.c.f.}}{f} \times c = 350 + \frac{57.5 - 55}{55} \times 50 = 352.27$$

$Q_3 = \frac{N}{4} \times 3 = \frac{230}{4} \times 3 = 172.5$ th item, which lies in the class 450-500.

$$\therefore Q_3 = L + \frac{\frac{N}{4} \times 3 - \text{p.c.f.}}{f} \times c = 450 + \frac{172.5 - 110}{67} \times 50 = 496.64$$

$\therefore$ Central 50% = Q_1 to Q_3

$$= 352.27 \text{ to } 496.64 \text{ [Ans.]}$$

(iii) The no. of persons having income 530 and above = $(39 \times \frac{20}{50}) + 14 = 29.6$

The required percentage of employees whose are taxable

$$= \frac{29.6}{230} \times 100$$

$$= 12.87\% \text{ [Ans.]}$$

Problem 3

Given below is the frequency distribution of the marks obtained by 90 students. Compute the arithmetic mean, median and mode:

Marks	No. of Students	Marks	No. of Students
20 – 29	02	60 – 69	18
30 – 39	12	70 – 79	10
40 – 49	15	80 – 89	09
50 – 59	20	90 – 99	04

Solution:

Calculation of Mean, Median and Mode

Marks	f	M.P.(X)	fX	c.f
20 – 29	2	24.5	49	2
30 – 39	12	34.5	414	14
40 – 49	15	44.5	667.5	29
50 – 59	20	54.5	1090	49
60 – 69	18	64.5	1161	67
70 – 79	10	74.5	745	77
80 – 89	9	84.5	760.6	86
90 – 99	4	94.5	378	90
	N = 90		ΣfX = 5265	

$$(i) \text{ Mean} = \frac{\Sigma fX}{N} = \frac{5265}{90} = 58.5$$

(ii) Median = Size of $\frac{N}{2}$ th item = $\frac{90}{2} = 45$ th item, which lies in the class 50 – 59. But the real class limit is 49.5 – 59.5.

$$\text{Median} = L + \frac{\frac{N}{2} - \text{pcf}}{f} \times i = 49.5 + \frac{45 - 29}{20} \times 10 = 57.5$$

(iii) For highest frequency 20, mode lies in the class 50 – 59. But the real class limit is 49.5 – 59.5.

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 49.5 + \frac{5}{5+2} \times 10 = 56.64$$

Ans: (i) Mean = 58.5, (ii) Median = 57.5, (ii) Mode = 56.64.

Problem 4

From the following table showing the wage distribution in a certain factory, determine:

(i) the mean wage; (ii) the median wage; (iii) the modal wage; (iv) the middle 50% of the wage earners.

Weekly wages (Tk)	No. of employees
1200 – 1400	8
1400 – 1600	12
1600 – 1800	20
1800 – 2000	30
2000 – 2200	40
2200 – 2400	32
2400 – 2600	18
2600 – 2800	7
2800 – 3000	6
3000 – 3200	4

Solution:

Calculation of Mean, Median and Mode

Weekly wage	f	X (m.p)	fX	c.f
1200 – 1400	8	1300	10400	8
1400 – 1600	12	1500	18000	20
1600 – 1800	20	1700	34000	40

1800 – 2000	30	1900	57000	70
2000 – 2200	40	2100	84000	110
2200 – 2400	32	2300	73600	142
2400 – 2600	18	2500	45000	160
2600 – 2800	7	2700	18900	167
2800 – 3000	6	2900	17400	173
3000 – 3200	4	3100	12400	177
	N = 177		$\Sigma fX = 370700$	

(i) $\bar{X} = \frac{\Sigma fX}{N} = \frac{370700}{177} = 2094.4$ [Ans.]

(ii) Median = Size of $\frac{N}{2}$ th item = $\frac{177}{2} = 88.5$ th item, which lies in the class 2000 – 2200.

$$\therefore \text{Median} = L + \frac{\frac{N}{2} - pcf}{f} \times i = 2000 + \frac{88.5 - 70}{40} \times 20 = 2092.5 \quad [\text{Ans.}]$$

(iii) For highest frequency 40, mode lies in the class 2000 – 2200.

$$\therefore \text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 2000 + \frac{10}{10 + 8} \times 200 = 2111.11 \quad [\text{Ans.}]$$

(iv) The limits for the middle 50% of the wage earners shall be given by Q_1 and Q_2

$Q_1 = \text{Size of } \frac{N}{4} \text{th item} = \frac{177}{4} = 44.25 \text{th item, which lies in the class 1800 – 2000.}$

$$\therefore Q_1 = L + \frac{\frac{N}{4} - pcf}{f} \times i = 1800 + \frac{44.25 - 40}{30} \times 200 = 1823.33$$

$Q_3 = \text{Size of } \frac{3N}{4} \text{th item} = 3 \times \frac{177}{4} = 132.75 \text{th item, which lies in the class 2200 – 2400.}$

$$\therefore Q_3 = L + \frac{\frac{3N}{4} - pcf}{f} \times i = 2200 + \frac{132.75 - 110}{32} \times 200 = 2342.2$$

∴ Required limits for the middle 50% = 1823.3 to 2342.2. [Ans.]

Problem 5

The following table gives the daily wages (in taka) of a certain tea garden:

Daily wages	30-32	32-34	34-36	36-38	38-40	40-42	42-44	44-46	46-48	48-50
No. of workers	2	9	25	30	49	62	39	20	11	3

- Determine the mean wages.
- Determine median wages.
- Determine modal wages.
- Determine the number of wage earners receiving between Tk 37 and Tk 45.

Solution:

Calculation of Mean, Median and Mode

Weekly wage	f	X (m.p)	fX	c.f
30 – 32	2	31	62	2
32 – 34	9	33	297	11
34 – 36	25	35	875	36
36 – 38	30	37	1110	66
38 – 40	49	39	1911	115
40 – 42	62	41	2542	177
42 – 44	39	43	1677	216
44 – 46	20	45	900	236
46 – 48	11	47	517	247
48 – 50	3	49	147	250
	N = 250		ΣfX = 10038	

(i) Mean, $\bar{X} = \frac{\Sigma fX}{N} = \frac{10038}{250} = 40.152$ [Ans.]

(ii) Median = Size of $\frac{N}{2}$ th item = $\frac{250}{2} = 125$ th item, which lies in the class 40 – 42.

$$\text{Median} = L + \frac{\frac{N}{2} - pcf}{f} \times i = 40 + \frac{125 - 115}{62} \times 2 = 40.32 \text{ [Ans.]}$$

(iii) For highest frequency 62, mode lies in the class 40 – 42.

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i = 40 + \frac{13}{13 + 23} \times 2 = 40.72 \text{ [Ans.]}$$

(iv) Required Determine the number of wage earners receiving between Tk 37 and Tk 45
 $= (30 \times \frac{1}{2}) + 49 + 62 + 39 + (20 \times \frac{1}{2}) = 175$ [Ans.]

Problem 6

In an examination of 675 candidates the examiner supplied the following information:

Marks Obtained	No. of Candidates	Marks Obtained	No. of Candidates
Less than 10%	7	Less than 50%	381
Less than 20%	39	Less than 60%	545
Less than 30%	95	Less than 70%	631
Less than 40%	201	Less than 80%	675

- (i) Calculate mean, median and mode of the percentage marks obtained.
- (ii) Obtain the limit of marks of central 50% of the candidates.

Solution:

Calculation of Mean, Median and Mode

Weekly wage	f	X (m.p)	fX	c.f
0% – 10%	7	5	35	7
10% – 20%	32	15	480	39
20% – 30%	56	25	1400	95
30% – 40%	106	35	3710	201
40% – 50%	180	45	8100	381
50% – 60%	164	55	9020	545
60% – 70%	86	65	5590	631
70% – 80%	44	75	3300	675
	N = 675		$\Sigma fX = 31635$	

(i) Mean, $\bar{X} = \frac{\Sigma fX}{N} = \frac{31635}{675} = 46.87$ [Ans.]

(ii) Median = Size of $\frac{N}{2}$ th item = $\frac{675}{2} = 337.5$ th item, which lies in the class 40 – 50.

$$\text{Median} = L + \frac{\frac{N}{2} - \text{pcf}}{f} \times i = 40 + \frac{337.5 - 201}{180} \times 10 = 47.58 \text{ [Ans.]}$$

(iii) For highest frequency 180, mode lies in the class 40 – 50.

$$\begin{aligned} \text{Mode} &= L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i \\ &= 40 + \frac{74}{74 + 16} \times 10 \\ &= 48.22 \text{ [Ans.]} \end{aligned}$$

Problem 7

An incomplete frequency distribution is as follows, where median is 65:

Variables	Frequency
30 – 40	10
40 – 50	15
50 – 60	?
60 – 70	20
70 – 80	?
80 – 90	5
	100

Find out the missing frequencies.

Solution: **Calculation of Missing Frequencies**

Variables	f	c.f
30 – 40	10	10
40 – 50	15	25
50 – 60	f_1	$25 + f_1$
60 – 70	20	$45 + f_1$
70 – 80	F_2	$45 + f_1 + f_2$
80 – 90	5	$50 + f_1 + f_2$
	$N = 100$	

We know, $\Sigma f = N$

$$\Rightarrow 50 + f_1 + f_2 = 100$$

$$\Rightarrow f_1 + f_2 = 50$$

$$\Rightarrow f_2 = 50 - f_1 \dots\dots\dots(1)$$

$$\therefore \text{Median} = L + \frac{\frac{N}{2} - \text{pcf}}{f} \times i$$

$$\Rightarrow 65 = 60 + \frac{50 - (25 + f_1)}{20} \times 10$$

$$\Rightarrow 5 = \frac{50 - 25 - f_1}{2}$$

$$\Rightarrow 10 = 25 - f_1$$

$$\Rightarrow f_1 = 15$$

$$(1) \Rightarrow f_2 = 50 - f_1$$

$$= 50 - 15$$

$$= 35$$

$$\therefore f_1 = 15, f_2 = 35 \quad [\text{Ans.}]$$

Problem 8

The expenditure of 1000 families is given as under:

Expenditure (Taka in thousand)	40 – 59	60 – 79	80 – 99	100 – 119	120 – 139
No. of families	50	?	500	?	50

Median and mean of the distribution are Tk 87.50 each. Calculate the missing frequencies f_1 and f_2 .

Solution:

Calculation of Missing Frequencies

Output (in units)	f	cf
40 – 59	50	50
60 – 79	f_1	$50 + f_1$
80 – 99	500	$550 + f_1$
100 – 119	f_2	$550 + f_1 + f_2$
120 – 139	50	$600 + f_1 + f_2$
	N = 1000	

Median = $\frac{N}{2} = \frac{1000}{2} = 500$ th item, which lies in the class 80-99. But the real class limit is 79.5-99.5.

$$\therefore \text{Median} = L + \frac{\frac{N}{2} - \text{p.c.f.}}{f} \times c$$

$$\Rightarrow 87.50 = 79.5 + \frac{500 - (50 + f_1)}{500} \times 20$$

$$\Rightarrow 87.50 - 79.5 = \frac{500 - 50 - f_1}{25}$$

$$\Rightarrow 8 = \frac{450 - f_1}{25}$$

$$\Rightarrow 8 \times 25 = 450 - f_1$$

$$\Rightarrow 200 = 450 - f_1$$

$$\Rightarrow f_1 = 450 - 200$$

$$\Rightarrow f_1 = 250$$

$$\therefore f_2 = 1000 - (50 + 250 + 500 + 50) = 150$$

$$\therefore f_1 = 250, f_2 = 150 \quad [\text{Ans.}]$$

.....

.....

Questions

1. What do you mean by measures of central tendency or location? Write different measurements of measures of central tendency.
2. Mention the desirable qualities of a good measure of central tendency.
3. What are the objectives of central tendency?
4. Define arithmetic mean, median and mode.
5. What are the characteristics of arithmetic mean?
6. Briefly discuss the arithmetic mean and median. Why arithmetic mean is so popular?
7. Distinguish between simple and weighted averages.
8. When geometric mean is preferred to arithmetic mean?
9. State the uses of Geometric average.
10. What are the usual measures of central tendency? Which of them do you prefer and why?
11. Discuss the relationship among mean, median and mode.
12. Define weighted arithmetic mean. Discuss the importance of weighted arithmetic mean in statistics.
13. Construct the relationship among Arithmetic mean, Geometric mean and Harmonic mean.
14. Prove that $AM \geq GM \geq HM$.

Problem: 1

The monthly income (in Tk) of 10 employees working in a firm is as follows:

4487 4493 4502 4446 4475 4492 4572 4516 4468 4489

Calculate the average monthly income. (4494)

Problem:2

The annual rates of growth of output of a factory in 5 years are 5.0, 7.5, 2.5, 5.0 and 10.0 respectively. What is the compound rate of growth (GM) of output per annum for the period?(2.328)

Problem:3

Three groups of observations contain 8, 7, and 5 observations. Their geometric means are 8.52, 10.12 and 7.75 respectively. Find the geometric mean of the 20 observations in the single group formed by pooling the three groups. (9.93)

Problem:4

A machine was purchased for Tk 10 lakh in 2013. Depreciation on the diminishing balance was charged @ 40% in the first year, 25% in the second year and 10% per annum during the next three years. What is the average depreciation charge during the whole period?(79.98)

Problem : 5

Calculate harmonic mean of numbers 10, 20, 25, 40, 50.(21.277)

Problem: 6

In a certain factory a unit of work is completed by A in 4 minutes, by B in 5 minutes, by C in 6 minutes, by D in 10 minutes and by E in 12 minutes.

- (a) What is the average number of units of work completed per minute?(6.25)
- (b) At this rate how many units will they complete in a six-hour day?(2250)

Problem: 7

A toy factory has assigned a group of 4 workers to complete an order of 1400 toys of a certain type. The productive rates of the four workers are given below:

Workers	Productive rates
A	4 minutes per toy
B	6 minutes per toy
C	10 minutes per toy
D	15 minutes per toy

Find the average minutes per toy by the group of workers.(6.861)

Problem: 8

The following are the figures of profits earned by 1400 companies during 2012-13.

Profit (Tk in lakh)	No. of companies	Profit (Tk in lakh)	No. of companies
---------------------	------------------	---------------------	------------------

200 – 400	500	1000 – 1200	100
400 – 600	300	1200 – 1400	80
600 – 800	280	1400 – 1600	20
800 – 1000	120		

Calculate the arithmetic mean (A.M), geometric mean(G.M) and harmonic mean (H.M) and find a relation.(AM=605.71 ,GM=532.11, HM=472.97)

Problem: 9

1500 workers are working in an industrial establishment. Their age is classified as follows:

Age (years)	No. of workers	Age (years)	No. of workers
18 – 22	120	38 – 42	184
22 – 26	125	42 – 46	162
26 – 30	280	46 – 50	86
30 – 34	260	50 – 54	75
34 – 38	155	54 – 58	53

Calculate the median age. Also show graphically. (33.461)

Problem:10

The profits earned by 100 companies during 2012-13 are given below:

Profits (Tk in lakh)	No. of companies	Profits (Tk in lakh)	No. of companies
20 – 30	4	60 – 70	15
30 – 40	8	70 – 80	10
40 – 50	18	80 – 90	8
50 – 60	30	90 – 100	7

Calculate Q_1 , median, D_4 and P_{80} and interpret the values.(47.22,56.67, 53.33,75)

Problem: 11

The following data relate to the sales of 100 companies:

Sales (Tk in lakh)	No. of companies	Sales (Tk in lakh)	No. of companies
Below 60	12	66 – 68	10
60 – 62	18	68 – 70	3
62 – 64	25	70 – 72	2
64 – 36	30		

Calculate the value of modal sales.(64.4)

Problem:12

The following data relate to the sales of 100 companies:

Sales (Tk in lakh)	No. of companies	Sales (Tk in lakh)	No. of companies
0 – 100	12	300 – 400	20
100 – 200	18	400 – 500	17
200 – 300	25	500 – 600	6

Draw the histogram and hence find the modal value. Check this value by direct calculation.

(258.33)

Problem 13

In a moderately asymmetrical distribution the mode and the mean are 32.1 and 35.4 respectively. Calculate the median. (34.3)

Problem:14

There are two branches of a company employing 100 and 80 employees respectively. If arithmetic mean of the monthly salaries paid by two branches is Tk 4570 and Tk 6750 respectively, find the arithmetic mean of the salaries of the employees of the company as a whole.(5538.89)

Problem:15

The mean of marks in statistics of 100 students of a class was 72. The mean of marks of boys was 75, while their member was 70. Find out the mean marks of girls in the class.(65)

Problem: 16

The mean age of a combined group of men and women is 30 years. If mean age of the group of men is 32 and that of the group of women is 25, find out the percentage of men and women in the group.(71.43%,28.57%)

Problem:17

The average weekly wage for a group of 25 persons working in a factory was calculated to be Tk 378.40. It was later discovered that one figure was misread as 160 instead of the correct value Tk 200. Calculate correct average wage.(380)

Problem: 18

Calculate mean, median and mode for the following data pertaining to marks in statistics out of 140 marks for 80 students in a class:

Marks more than	0	20	40	60	80	100	120
No. of students	80	76	50	28	18	9	3

Problem: 19

The following table gives the distribution of weekly wages of 600 workers of a factory:

Weekly wages (in Tk)	Frequency	Weekly wages (in Tk)	Frequency
Below 375	69	600 – 675	58
375 – 450	167	675 – 750	24
450 – 525	207	750 – 825	10
525 – 600	65		

Find mean, median and mode. **(486,473.188,466.48)**

Problem: 20

From the following data find out the missing frequency when mean is 30.

Central Value 10 20 30 40 50

Frequency 5 10 15 ? 5

(10)

Problem: 21

Find the missing frequencies in the following distribution if N is 100, and median 30:

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No of students	10	15	?	30	10	?

(25,10)

.....